

Continuum-Mechanics-MFEM

Verification Manual (`apps/` and `tests/`)

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Abstract

This manual documents how the flux-kernel framework of the root `src/` library is verified. It describes the application executable and the scripts under `apps/`, the example meshes, and every input under `apps/input/`: the problem it poses, the reference it is compared with, the quantity checked and the tolerance, and the test that runs it. The unit-level checks of the test suite (`tests/`) are summarised first, since they verify the ingredients the application cases rely on. The models and methods themselves are described in the theory manual (`doc/theory_manual.tex`); the closed-form solutions of the homogeneous states of the incompressible models are collected in Appendix A.

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1 Organisation

1.1 Levels of verification

The checks form three tiers, each relying on the previous one.

- Unit and consistency tests** (tests/test_base, test_materials): tensor algebra against hand values, dual derivatives against finite differences, the YAML schema and the expression parser, and every material's stress-free reference state, tangent, objectivity and small-strain limit.
- Discretisation tests** (tests/test_solid_mms, test_mixed, test_homogeneous, test_loading, test_linear_elasticity): patch tests, assembled Jacobians against finite differences, manufactured solutions with measured convergence rates, homogeneous states against closed forms, and the boundary-condition and load-scheduling features.

3. **Application cases** (apps/input/, driven by tests/test_verification, test_linear_verification, test_benchmarks, test_loading, test_parallel and apps/homogeneous_compare.py): complete inputs run through the same library wiring as the executable and compared with analytical solutions, literature benchmarks, frozen regression values, or a reference code.

1.2 Running the suite

Target	Content
make	build/libcmf.a, build/apps/solid_mechanics, build/tests/*
make meshes	regenerates apps/mesh/*.msh from the .geo sources with Gmsh (the .msh files are kept in the tree)
make check	serial, about 45 s: test_base, test_materials, test_solid_mms, test_mixed, test_homogeneous, test_loading, test_linear_elasticity, test_dynamics
make homogeneous	the executable on the homogeneous-deformation inputs, compared with the closed forms by apps/homogeneous_compare.py
make elastic_bar	the elastic bar exercise (Section 8): table and plot of force against displacement by apps/elastic_bar_compare.py
make plate_with_hole	the plate-with-a-hole exercise (Section 9): errors against Kirsch's closed form and plot by apps/plate_with_hole_compare.py
make dynamics	the dynamic cases (Section 7): time histories over their closed forms, measures and plots by apps/dynamics_compare.py (about two and a half minutes)
make test	everything: check, the Cook inputs (finite strain and small strain, both formulations) serial and with four ranks, test_linear_verification (about 30 s), test_dynamic_verification (about two minutes, then its parallel consistency part with two and four ranks) and dynamics_compare.py -check, parallel consistency with two and four ranks, the homogeneous comparison, test_mixed -full, test_benchmarks serial and parallel, test_loading with four ranks, test_verification, and finally the Cook ratio gate

Targets are run from the repository root, since inputs are referenced as apps/input/<set>/<case>.yaml. Every test returns a non-zero exit status on the first failed check and prints the measured quantities next to their references.

1.3 The executable

apps/solid_mechanics.cpp contains no physics: it parses the input, builds the parallel mesh, creates the problem selected by **formulation**, installs the loads, and runs the quasi-static stepper with a callback that writes ParaView cycles, probes and reactions after every converged increment. With a **dynamics** block in the input it wraps the same problem with inertia and runs the same loop in physical time; the header then lists the scheme, the time steps and the time dependence of every load and Dirichlet entry, the per-step lines gain the energies (**output.energy**), and the reactions include the inertia a support carries. At startup it prints the mesh statistics and the element and boundary attributes with their names; at the end the Newton report, $\|\mathbf{u}\|_{L^2}$, the internal energy, the probes and the reactions. The exit status is 0 on convergence, 2 if the load path did not complete, and 1 on an input error. Run as

```
./build/apps/solid_mechanics -i apps/input/finite_elasticity/cooks_membrane/cook.yaml
mpirun -np 4 ./build/apps/solid_mechanics -i <input.yaml>
```

1.4 Meshes

Every input reads a Gmsh mesh from `apps/mesh/`, generated from a `.geo` source with named physical groups that the boundary conditions refer to. The box-like geometries keep MFEM's Cartesian numbering (2D: bottom 1, right 2, top 3, left 4; 3D: bottom $z=0$ 1, front $y=0$ 2, right $x=L$ 3, back $y=L$ 4, left $x=0$ 5, top $z=L$ 6).

Source	Meshes	Geometry and groups
<code>square.geo</code>	<code>square.msh</code>	unit square, $n \times n$ quadrilaterals; bottom, right, top, left
<code>cook.geo</code>	<code>cook.msh</code>	Cook's membrane, transfinite quadrilaterals (the bilinear map of the former Cartesian input); bottom, right, top, left
<code>box.geo</code>	<code>cube</code> , <code>beam</code> , <code>cube10</code> , <code>shear_cube</code> , <code>column_*</code>	$L_x \times L_y \times L_z$ hexahedra; bottom, front, right, back, left, top
<code>cube_tet.geo</code>	<code>cube_tet.msh</code>	unit cube in tetrahedra, same groups
<code>annulus.geo</code>	<code>annulus.msh</code>	quarter annulus $A=1$, $B=2$, second order; bottom, outer, left, inner
<code>cylinder_torsion.geo</code>	<code>cylinder_torsion.msh</code>	solid cylinder $a=12.7$, $L=25.4$, second-order tetrahedra; bottom, top, wall
<code>plate_hole.geo</code>	<code>plate_hole.msh</code>	quarter plate $15 \times 10 \times 1$ with a 3 mm hole, second order; xbot, ybot, xtop, ytop, hole, zbot, ztop
<code>plate_hole_2d.geo</code>	<code>plate_hole_2d.msh</code>	quarter of a 40×40 plate with a unit hole, second-order quadrilaterals
<code>tube_quarter.geo</code>	<code>tube_quarter.msh</code>	quarter tube $10/11 \times 5$, second order; xplane, yplane, zbot, ztop, inner, outer
<code>sphere_octant.geo</code>	<code>sphere_octant.msh</code>	octant shell $10/11$, second order; xplane, yplane, zplane, inner, outer
<code>footing.geo</code>	<code>footing.msh</code>	50 mm cube with a quarter patch on top; left, front, bottom, patch, top_rest, right, back
<code>inclusion.geo</code>	<code>inclusion.msh</code>	octant of a cube with a spherical inclusion; volumes inclusion, matrix; the six box faces
<code>perturb_column.py</code>	<code>column_buckling</code> , <code>column_euler</code>	adds the imperfection $x, y \rightarrow \delta_0(1 - \cos 2\pi z/L)$ to a straight box mesh

2 Unit and consistency tests

2.1 test_base

Tensor algebra (determinant, inverse, contractions of rank 2 and 4) against hand values in two and three dimensions; dual-number derivatives of scalar composites and of det, inv and tensor compositions against Richardson-extrapolated central differences to 10^{-10} ; the YAML schema: the documented example parses to the expected values, defaults for omitted sections, malformed files give an error rather than a crash, missing keys, wrong types, unknown keys and bad enumerations name the full key path; boundary attributes by number and by physical-group name; load schedules and their YAML forms (ramp, constant, table, components, step segments, bisection control); the expression grammar (precedence, right-associative power, functions, comparisons, parse errors with column numbers); and material regions (inheritance, bulk-key replacement, errors).

2.2 test_materials

For every model, at five random deformation gradients $\mathbf{F} = \mathbf{I} + 0.4\mathbf{R}$ with $\det \mathbf{F} > 0.3$:

- stress-free reference state $\mathbf{P}(\mathbf{I}) = \mathbf{0}$ to machine precision;
- the dual-number tangent against central finite differences of $\mathbf{P}$ (step 10^{-6}), relative error $\leq 10^{-6}$;
- objectivity $\mathbf{P}(\mathbf{QF}) = \mathbf{QP}(\mathbf{F})$ for random rotations $\mathbf{Q}$ to 10^{-12} ;
- the small-strain limit: for $\mathbf{F} = \mathbf{I} + \epsilon\mathbf{H}$ the stress equals $\lambda \operatorname{tr} \epsilon \mathbf{I} + 2\mu\epsilon$ with the resolved moduli ($\lambda = \kappa - \frac{2}{3}\mu$ for the decoupled models);
- for the decoupled models, $\mathbf{P} = \partial\Psi/\partial\mathbf{F}$ and $\mathbf{P}_{\text{iso}} = \partial\Psi_{\text{iso}}/\partial\mathbf{F}$ by duals against finite differences, and the deviatoric property $\operatorname{tr}(\mathbf{P}_{\text{iso}}\mathbf{F}^T) = 0$ at every $\mathbf{F}$;
- Ogden reduces to the isochoric neo-Hookean model for $(\mu, \alpha) = (\mu, 2)$ and to Mooney–Rivlin for $(2c_1, 2), (-2c_2, -2)$; its analytic tangent matches finite differences also at coincident principal stretches;
- the plane-stress adapter: $P_{33} = 0$ and no out-of-plane shear stress, $J = 1$ for incompressible bases, the in-plane dual tangent against finite differences (the compressible branch differentiates a converged root), $\partial\Psi/\partial\mathbf{F} = \mathbf{P}$ in the plane (envelope theorem), objectivity under in-plane rotations, the small-strain plane-stress moduli $E/(1-\nu^2)$, $E\nu/(1-\nu^2)$ and the thickness strain $-\nu/(1-\nu)\epsilon$, and uniaxial tension of an incompressible sheet, $P_{11} = \mu(\lambda - \lambda^{-2})$;
- the volumetric laws (quadratic, Simo–Taylor, logarithmic, $J \ln J$): $u(1) = u'(1) = 0$, $u''(1) = 1$, u' and u'' against dual and finite-difference derivatives of u and u' , and the isochoric neo-Hookean and Ogden models through the full decoupled checks above with each law; the factory applies the key and the coupled models refuse it;
- resolution of (μ, κ) from every admissible key combination and the refusal of the inadmissible ones;
- **linear_elastic**: the checks above without objectivity (the model is not objective, by construction) and with the small-strain limit taken at a strain of 5%, where it holds to 10^{-14} because the model is that limit (at the generic $\epsilon = 10^{-7}$ it would sit on the $\epsilon_{\text{mach}}/|\operatorname{Grad} \mathbf{u}|$ cancellation floor, 8×10^{-10}); $\mathbf{P}(\mathbf{I} + \mathbf{W}) = \mathbf{0}$ for skew $\mathbf{W}$; the dual tangent equal to the closed-form $\mathbb{C}$ at two random $\mathbf{F}$ (constant, minor and major symmetries); $\Psi = \frac{1}{2}\boldsymbol{\sigma} : \boldsymbol{\epsilon}$; the split $\mathbf{P} = \mathbf{P}_{\text{iso}} + \kappa \operatorname{tr} \epsilon \mathbf{I}$ with a deviatoric $\mathbf{P}_{\text{iso}}$; under plane stress, stress, thickness strain and tangent against the closed forms at a finite strain, for $\nu = 0.3$ ($\lambda^* = 2\lambda\mu/(\lambda + 2\mu)$) and $\nu = \frac{1}{2}$ ($\lambda^* = 2\mu$); the factory and its key rules.

3 Discretisation tests

3.1 Patch tests, manufactured solutions, Jacobians (test_solid_mms)

On a perturbed quadrilateral mesh an affine displacement $\mathbf{u} = \mathbf{A}\mathbf{X} + \mathbf{c}$ prescribed on the boundary is reproduced exactly (patch test) for both coupled models. The smooth manufactured field $\mathbf{u} = \alpha(\sin \pi X \sin \pi Y, X^2 Y(1 - Y))$ with the body force $\mathbf{b} = -\rho_R^{-1} \operatorname{Div} \mathbf{P}(\mathbf{F}(\mathbf{u}))$ (divergence by central differences of the closed-form stress) gives L^2 convergence rates of at least $k + 0.9$ for $k = 1, 2$; on the finest $k = 2$ mesh the last Newton steps above the round-off floor contract quadratically (order estimate ≥ 1.5); the assembled Jacobian matches central differences of the residual along a random direction at a nonzero state.

3.2 Mixed formulation (test_mixed)

Affine patch tests with a constant pressure $p_0 = U'(\det(\mathbf{I} + \mathbf{A}))$ for the decoupled materials at $\nu = 0.45$ and at $\kappa = \infty$, and at $\nu = 0.45$ with the Simo–Taylor, logarithmic and $J \ln J$ volumetric laws (exact to 10^{-14} , quadratic Newton convergence); the block Jacobian against finite differences (4×10^{-9}), with the pressure rows checked separately so that the law-dependent block K_{pu} is exercised; manufactured solutions with a volume-preserving motion (the composition of two shears $x = X + a \sin \pi Y$, $y = Y + b \sin \pi x$, $J = 1$, with the smooth pressure $p = c \sin \pi X \cos \pi Y$) for the incompressible case, rates $u \geq 3.0$, $p \geq 2.0$, and a compressible field at $\nu = 0.45$ with $p = \kappa(J - 1)$, rates $u \geq 3.4$, $p \geq 2.0$; and agreement of the mixed and displacement formulations at finite κ under refinement. The pressure is made unique by a manufactured traction on one edge. -full adds the 32×32 levels.

The same gates for `linear_elastic` (Herrmann’s formulation): the mixed stress at $p = \kappa \operatorname{tr} \boldsymbol{\varepsilon}$ equals the displacement formulation’s; patch tests at an amplitude of 0.3 with $p_0 = \kappa \operatorname{tr} \mathbf{A}$, and with a traceless $\mathbf{A}$ and a free p_0 at $\kappa = \infty$, exact in two Newton steps (the first reaches $4 \times 10^{-11} \|R_0\|$, the second refines); a block Jacobian that does not depend on the state (difference exactly zero) and is symmetric (4×10^{-17}); manufactured solutions at $\nu = 0.45$ with $p = \kappa \operatorname{div} \mathbf{u}$ at an amplitude of 0.2 ($|\operatorname{Grad} \mathbf{u}| \approx 0.6$) and at $\nu = \frac{1}{2}$ with a divergence-free field from the stream function $\psi = \sin \pi X \sin \pi Y / \pi$, rates $u \geq 3.0$, $p \geq 2.0$; agreement with the displacement formulation under refinement; follower pressures, the tangent predictor and $\nu = \frac{1}{2}$ in the displacement formulation refused.

3.3 Homogeneous deformations (test_homogeneous)

For the six incompressible models with rubber-like parameters (isochoric neo-Hookean $\mu = 1$; Mooney–Rivlin $c_1 = 0.4$, $c_2 = 0.1$; Yeoh 0.5, $-0.05, 0.01$; Gent $\mu = 1$, $J_m = 10$; Arruda–Boyce $\mu = 1$, $N = 5$; Ogden (0.63, 0.0012, -0.01), (1.3, 5.0, -2.0)):

1. at the material point, the code’s Cauchy stress at $\mathbf{F} = \operatorname{diag}(\lambda_a)$ with the analytic mean stress equals the analytic principal stresses of the plane-strain, uniaxial, equibiaxial and pure-shear states, and the deviatoric stress in simple shear equals its spectral closed form with $\sigma_{11} - \sigma_{22} = \gamma \sigma_{12}$;
2. as finite element solutions of the mixed formulation on 4×4 (plane strain) and $2 \times 2 \times 2$ (uniaxial) Q2/Q1 meshes, and of the plane-stress displacement formulation for the sheet states uniaxial, equibiaxial, pure shear and simple shear on a 4×4 Q2 mesh: displacement, pressure and every quadrature quantity (Cauchy and first Piola–Kirchhoff stress, J , von Mises, energy density, thickness stretch) in all presentations (both nodal projections, element averages, raw quadrature values), exact to solver tolerance with quadratic Newton convergence.

3.4 Boundary conditions and loading (test_loading)

On Cook’s membrane and unit boxes: staged loading (a traction ramped over $[0, 0.5]$ and held, a second ramped over $[0.5, 1]$) equals the single-stage solve, and the state at $t = 0.5$ equals the first traction alone (path independence); a load-unload table returns to zero; with a capped Newton the planned step fails and bisection recovers, while without substepping the failure is reported; the tangent predictor reaches the same converged state as the default start with fewer Newton iterations (Cook’s membrane with a prescribed edge, 18 to 9; a clamped sheared square in the mixed formulation with the logarithmic volumetric law at $\kappa/\mu = 1000$, 17 with a damped step to 6 full steps) and leaves a traction-driven path unchanged; a programmatic time-dependent coefficient with the constant schedule reproduces the ramp; the manufactured problem written entirely as

YAML expressions converges at rate ≥ 2.8 for $k = 2$, also with only the x component prescribed on two faces and the exact traction supplied for the other; a traction expression linear in t equals value plus ramp and one without t is assembled once; the uniaxial symmetry cube with rollers reproduces the affine field and closed-form pressure of the full cube, with exactly the expected number of essential degrees of freedom; rollers on all six faces under a body force leave no rigid mode; the follower tangent matches central differences on distorted quadrilateral and hexahedral meshes at a random state; the follower pressure equals the dead pressure at $\mathbf{F} = \mathbf{I}$ and agrees to $O(p^2)$ under a small load; the thick-walled cylinder inflates as Rivlin says (Section 4.4); two-region materials reproduce the single-material run when the parameters agree, have a consistent tangent and deform less in the stiff half; and the reactions equal P_{11} times the face area on the uniaxial cube and balance the applied resultant on Cook's membrane to 10^{-14} .

3.5 Small-strain linear elasticity (`test_linear_elasticity`)

The total Lagrangian kernel with the flux $\boldsymbol{\sigma}(\text{sym}(\mathbf{F} - \mathbf{I}))$ must be the classical linear elasticity operator, and the solve must be linear.

- *Oracle.* On perturbed quadrilateral, triangular, hexahedral and tetrahedral meshes, $k = 1, 2$, the assembled Jacobian applied to a random vector equals a `ParBilinearForm` with MFEM's `ElasticityIntegrator` and the kernel's quadrature rule to 5×10^{-16} ; it equals the assembled Jacobians of `neo_hookean`, `st_venant_kirchhoff` and `iso_neo_hookean` at $\mathbf{u} = \mathbf{0}$ (bitwise here), with a control away from $\mathbf{u} = \mathbf{0}$, where they must differ by $O(|\text{Grad } \mathbf{u}|)$.
- *Linearity.* With Dirichlet data, a traction and a body force: one Newton iteration from $\mathbf{u} = \mathbf{0}$ and from a random start, and at a thousand times the load (strains of order one to ten), where the solution scales to 10^{-14} ; superposition of the three loads; Clapeyron's theorem $2\Psi_{\text{int}} = \mathbf{f}_{\text{ext}} \cdot \mathbf{u}$, all to 10^{-10} or better.
- *Patch tests* at an amplitude of 0.1 on the four element types: 10^{-14} in one iteration (the hyperelastic patch tests need amplitudes below 10^{-5} for that).
- *Manufactured solution* with the body force $-(\lambda + \mu) \text{grad div } \mathbf{u} + \mu \Delta \mathbf{u}$ written out, so that the check does not pass through the material's own stress function: L^2 rates 1.97–2.00 ($k = 1$) and 3.00 ($k = 2$).
- *Finite-strain limit.* `st_venant_kirchhoff` approaches it linearly in the load: the relative difference halves with the load (ratios 2.00).
- *Outputs.* Five homogeneous Hooke states at a strain of 5% (three-dimensional uniaxial stress and a general gradient with a rotation part, plane strain, plane stress uniaxial and general) solved by finite elements; every quantity in every presentation against closed forms built from E , ν and the gradient alone, to 6×10^{-13} of a stress level of 12.5; end-face reactions σA on the reference area. With the finite-strain push-forward left in, this test fails by 3%.
- *Reactions and input errors.* Force and moment balance of a clamped, loaded square with reference moment arms; follower pressures and the tangent predictor refused.
- *The floor of the residual.* A 20:1 beam whose exact first step leaves $\|R\|/\|R_0\| = 1.6 \times 10^{-9}$, above the default Newton tolerance: with $\text{rtol} = 10^{-15}$ it is accepted at the floor after two steps with the same solution, whereas `st_venant_kirchhoff` under the same tolerance still fails.
- *One Jacobian for a load path.* Ten steps with a ramped Dirichlet datum and a traction loaded and partly unloaded by a table schedule: one assembly and one AMG setup with the reuse, ten of each without, and the two paths identical bit for bit at every step (norm, sum and final vector compared with `==`); the same for the incompressible mixed problem over five steps (one saddle-point setup against five). After `ClearBoundaryConditions` and other boundary

conditions the Jacobian is reassembled once and the solution is that of a fresh problem. A `st_venant_kirchhoff` path assembles and sets up once per Newton iteration, as before.

3.6 Inertia and time integration (`test_dynamics`)

The dynamic analysis is a decorator over the quasi-static problem (theory manual, Section “Elastodynamics”): a constant mass matrix, the step equation $\mathbf{S}(\mathbf{u}) + c_M \mathbf{M}(\mathbf{u} - \mathbf{u}^*) + \mathbf{h}_n = \mathbf{0}$ with Jacobian $\mathbf{K} + c_M \mathbf{M}$, and the load stepper in physical time. 221 checks in seven seconds.

- *Mass.* On perturbed quadrilateral, triangular, hexahedral and tetrahedral meshes, $k = 1, 2$, with two densities by region: $\mathbf{1} \cdot \mathbf{M} \mathbf{1} = \sum_r \rho_r V_r$ to 2×10^{-16} (the reference volumes by a rule that is exact for the trilinear Jacobian), symmetry, no coupling between components, and the eliminated copy acting as $\mathbf{M}$ on the free degrees of freedom.
- *Free fall.* No supports, a body force $\mathbf{g}$, two densities, `linear_elastic` and `neo_hookean`, quadrilaterals and tetrahedra: the trapezoidal rule is exact for a constant acceleration, so $\mathbf{u} = \frac{1}{2} \mathbf{g} t^2$ at every node on any mesh. Measured: $\mathbf{a}_0 = \mathbf{g}$ to 4×10^{-14} (the step load is on at $t = 0$), $\mathbf{u}$ to 1.4×10^{-13} over 50 steps, no strain energy. The test fails at once if the mass and the body force disagree on the density of a region, which they did before this test existed: a region’s `rho0` was read and ignored.
- *Order in Δt without spatial error.* $\mathbf{u} = \sin(\omega t) \mathbf{U}(\mathbf{X})$ with $\mathbf{U}$ quadratic, which the $k = 2$ space on an affine mesh holds exactly, with tractions or with prescribed motion on the boundary: rates 1.999–2.000 and $2.05 \rightarrow 2.005$ for displacement and velocity, for `newmark`, `hht` (0.1) and `generalized_alpha` (0.8). The mesh is 2×2 elements and the steps resolve its every mode: the error at a fixed time carries the free vibrations that the truncation error excites, each with the phase of its numerical frequency, and on a finer mesh with coarser steps the ratios scatter between 1.1 and 3.2 while the errors still fall as Δt^2 .
- *Energy and numerical dissipation.* Linear free vibration: the trapezoidal rule keeps the energy to 3×10^{-14} over 500 steps; with $\rho_\infty = 0.8$ the energy never exceeds its initial value (it is not monotone: the scheme dissipates in a norm of its own). With $\omega \Delta t \gg 1$ for every mode, $\rho_\infty = 0$ leaves 10^{-32} of the energy after ten steps, $\rho_\infty = 0.5$ makes $\ln E_n$ fall with the slope -1.331 against $2 \ln \rho_\infty = -1.386$ (the roots at infinity are a defective $-\rho_\infty$, hence a slope and not a ratio), and $\rho_\infty = 1$ keeps it to twelve digits: the parameter map of the family.
- *Balances.* A step traction and a moving support, trapezoidal rule, linear: kinetic plus internal energy minus the external work of load and support is constant to 1.6×10^{-14} of the peak energy (an algebraic identity of the scheme), and the reactions plus the external force equal the resultant of $\mathbf{M} \mathbf{a}$ to 5×10^{-13} .
- *Nonlinear path.* Neo-Hookean block thrown to displacements of 0.75 of its size: Newton order 1.90 with $\mathbf{K} + c_M \mathbf{M}$, self-convergence ratios 3.989 and 3.982; a manufactured solution in space and time converges at the spatial rate 3.015 under joint refinement.
- *One Jacobian per run.* Linear problem, 200 steps: one assembly of $\mathbf{K}$, one of $\mathbf{K} + c_M \mathbf{M}$, one AMG setup, and not a digit different from 200 of each. A step made to fail once is halved and the run equals the one with the two half steps planned; $\mathbf{K} + c_M \mathbf{M}$ is then formed three times (for Δt , $\Delta t/2$ and Δt again). Without the external work the accepted step does not evaluate the full static residual; state and reactions are the same to the last digit.
- *Stepper.* $t_{\text{final}} = 10^{-3}$ and 10^3 with 1000 steps, one of them halved: 1001 accepted steps, the end time is t_{final} exactly, no degenerate step.
- *Input.* The bar input through the library: fields `velocity` and `acceleration` registered and probed, the tip displacement equal to $A \cos(\tilde{\omega} t)$ with the scheme’s own dispersion $\tan(\tilde{\omega} \Delta t/2) = \omega \Delta t/2$ to $10^{-9} A$. The parser (85 checks in `test_base`): every error of the

dynamics block, schedules in physical time, the default schedule of each analysis.

- *Mixed formulation.* A Herrmann problem whose displacement (quadratic) and pressure (linear) the Taylor–Hood pair holds exactly, at $\nu = 0.4999$ and incompressible: order 2.00 for $\mathbf{u}$ and 1.97 to 2.11 for p . The pressure carries the tolerance of the solves multiplied by $c_M = O(\Delta t^{-2})$, so the study uses tight solves and 50 to 400 steps; with the usual tolerances that noise (4×10^{-5}) hides the order of p from 400 steps on. Two incompressible runs that differ in p_0 alone differ at every later step by exactly ρ_∞ times the step before (0.6000, seven times): the pressure mode. At finite κ the initial pressure is $\kappa \operatorname{div} \mathbf{u}_0$ to 6×10^{-14} .

3.7 Parallel consistency (test_parallel)

The Cook inputs of both formulations, finite strain and small strain, are run serially with tightened tolerances (`-write`) and the recorded norms are matched to 10^{-10} relative by runs with two and four ranks (`-check`; the measured differences are 10^{-15}). A linear problem may be accepted at the round-off floor of its residual, just above the 10^{-12} asked for.

4 Verification cases (apps/input/finite_elasticity/verification/)

Every input in this directory is compared with an independent reference by a test. The header of each input states the reference and the formulas. Unless noted, the material parameters are dimensionless.

4.1 Homogeneous deformations of incompressible models

Problem. A unit square or cube carries an affine displacement $\mathbf{u} = \mathbf{G}\mathbf{X}$, $\mathbf{G} = \operatorname{diag}(\lambda_a - 1)$, on its constrained faces, written as expression strings (for example `["x", "-0.5*y"]`), with the remaining faces free; the isochoric neo-Hookean model with `incompressible: true` and $\mu = 1$ ($\mu = 0.5$ MPa in the three uniaxial inputs, which share their shear modulus with Section 4.2). The state is homogeneous, so $\operatorname{Div} \mathbf{P} = 0$ holds identically and the affine field with a constant pressure is the exact solution, which lies in the Taylor–Hood spaces.

Input	Formulation	Stretches	Constrained faces
plane_strain_neo_hookean	mixed, 2D	$(2, \frac{1}{2}, 1)$	left, right
plane_stress_neo_hookean	plane stress, 2D	$(2, 2^{-1/2}, 2^{-1/2})$	left, right
uniaxial_neo_hookean	mixed, 3D	$(2, 2^{-1/2}, 2^{-1/2})$	left, right
symmetry_uniaxial_neo_hookean	mixed, 3D	same, octant model	rollers on X, Y, Z on right
incompressible_uniaxial_iso_neo_hookean	mixed, 3D	$(1.5, 1.5^{-1/2}, 1.5^{-1/2})$, octant model	rollers on X, Y, Z on right
equibiaxial_neo_hookean	mixed, 3D	$(1.5, 1.5, 1.5^{-2})$	left, right, front, back
pure_shear_neo_hookean	mixed, 3D	$(1.5, 1, 1.5^{-1})$	left, right, front, back

Reference. The closed forms of Appendix A. With $\beta_a = \lambda_a \partial \Psi / \partial \lambda_a$ up to an isotropic term ($\beta_a = 2\Psi_1 \lambda_a^2 - 2\Psi_2 \lambda_a^{-2}$ for invariant models, $\sum_r \mu_r \lambda_a^{\alpha_r}$ for Ogden), the principal Cauchy stresses are $\sigma_a = \beta_a - \beta_f$ where f is the traction-free axis, $p = \frac{1}{3} \sum \sigma_a$, $\sigma_{vm} = \sqrt{\frac{3}{2} \sum (\sigma_a - p)^2}$, $P_a = \sigma_a / \lambda_a$. For the neo-Hookean model: uniaxial $\sigma_1 = \mu(\lambda^2 - \lambda^{-1})$; equibiaxial $\sigma_1 = \sigma_2 = \mu(\lambda^2 - \lambda^{-4})$; pure shear $\sigma_1 = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_2 = \mu(1 - \lambda^{-2})$; plane-strain extension $\sigma_{11} = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_{33} = \mu(1 - \lambda^{-2})$.

Check. `apps/homogeneous_compare.py` (`make homogeneous`) runs every input, reads the affine data and the constrained faces from the YAML to derive the stretches and the free axis, and compares the probed displacement, pressure or thickness stretch, von Mises stress, J and every component of $\boldsymbol{\sigma}$ and $\mathbf{P}$, in the nodal and the element presentations, with the closed forms; relative tolerance 10^{-8} (measured agreement about 10^{-13}). Any of the six models runs in the same inputs by changing the `material` line; the script's closed forms cover all of them.

4.2 Compressible uniaxial tension

Problem. The unit cube with rollers on $X, Y, Z = 0$ and the face $X = 1$ moved by $0.5t$ in x only (stretch $\lambda_1 = 1.5$), all other faces free. Six inputs with the same small-strain moduli, $\mu = 0.5$ MPa and $\nu = 0.45$ ($\kappa = 29/6$ MPa, $\kappa/\mu \approx 9.7$, compressible enough for the volumetric laws to differ): coupled neo-Hookean and Saint Venant–Kirchhoff with $E = 1.45$, $\nu = 0.45$ (displacement formulation), and the isochoric neo-Hookean model with $\mu = 0.5$, $\kappa = 29/6$ in the mixed formulation with each volumetric law: quadratic (`...mixed`), `simo_taylor`, `j_log_j` and `logarithmic` (the mixed constraint with $u'' \neq 1$). `incompressible_uniaxial_iso_neo_hookean` (Section 4.1) is their limit $\kappa = \infty$. A seventh input, `compressible_uniaxial_gent_compressible_summit`, uses SUMMIT's compressible Gent model with $\mu = 0.5$, $\kappa = 29/6$ and $J_m = 100$ (displacement formulation); its κ scales a quartic penalty and is not a small-strain bulk modulus, so that block is far more compressible than the others.

Reference. The state is homogeneous with $\mathbf{F} = \text{diag}(\lambda_1, \lambda_2, \lambda_2)$ and the lateral stretch fixed by $\sigma_{22} = 0$. The test solves $P_{22}(\text{diag}(\lambda_1, \lambda_2, \lambda_2)) = 0$ for λ_2 with the material's own stress function (which carries the volumetric law), then forms P_{11} , σ_{11} and J .

Check (`test_verification`). Axial displacement to 10^{-10} , lateral displacements to 10^{-8} , P_{11} , σ_{11} and J to 10^{-7} relative, and the lateral nominal stresses vanish to $10^{-7}P_{11}$.

Large stretch (`apps/uniaxial_plots.py`, `apps/neo_hookean_compare.py`). The first script drives every uniaxial input of this and the previous section to $\lambda_1 = 8$ in 70 load steps (Saint Venant–Kirchhoff to 0.95 of $\sqrt{1+1/\nu}$, where its lateral stretch $\lambda_2^2 = 1 - \nu(\lambda_1^2 - 1)$ vanishes), reads the ParaView series back and plots $P_{11}(\lambda_1)$ on an analytical curve coded independently of `src`: λ_2 by bisection of $P_{22} = 0$ with $P_{aa} = \mu(\lambda_a - \lambda_a^{-1}) + \lambda \ln J/\lambda_a$ (coupled neo-Hookean) or $P_{aa} = \mu J^{-2/3}(\lambda_a - I_1/(3\lambda_a)) + U'(J)J/\lambda_a$ (decoupled), the closed forms $\lambda_2^2 = 1 - \nu(\lambda_1^2 - 1)$, $P_{11} = E\lambda_1(\lambda_1^2 - 1)/2$ for Saint Venant–Kirchhoff, $P_{aa} = \mu J_m \lambda_a / (J_m - I_1 + 3) + (2\kappa A^3 (J^2 - 1) - \mu)/\lambda_a$ for the SUMMIT Gent model (which shows the Gent upturn towards its locking stretch, $I_1 - 3 = 61.7$ of $J_m = 100$ at $\lambda_1 = 8$), and $P_{11} = \mu(\lambda_1 - \lambda_1^{-2})$ when incompressible. Measured agreement is 3×10^{-12} relative or better at every step, and the nodal P_{11} is uniform to 2×10^{-11} . The second script overlays the neo-Hookean responses: at $\lambda_1 = 8$ the decoupled models lie 33% (quadratic) to 51% (logarithmic, $J = 2.9$, beyond the peak of $\ln J/J$ at $J = e$) below the incompressible curve, while the coupled model, for which $P_{22} = 0$ gives $P_{11} = \mu(\lambda_1 - \lambda_2^2/\lambda_1)$, stays on it.

4.3 Manufactured solutions

Problem. An exact displacement is prescribed on the whole boundary and the body force $\mathbf{b} = -\rho_R^{-1} \text{Div } \mathbf{P}(\mathbf{F}(\mathbf{u}))$ that makes it the solution is supplied as expression strings generated by sympy (expanded, no simplification). Saint Venant–Kirchhoff with $E = 250$, $\nu = 0.3$, $\rho_R = 2.5$ unless noted, $k = 2$.

Input	Mesh	Exact displacement
mms_2d_plane_strain	square.msh	$0.05 (\sin \pi x \sin \pi y, x^2 y (1 - y))$
mms_3d_hex	cube.msh	$0.05 (\sin \pi x \sin \pi y \sin \pi z, x^2 y (1 - y) z, x z (1 - z) \cos \frac{\pi y}{2})$
mms_3d_tet	cube_tet.msh	the same field on tetrahedra
mms_3d_mixed	cube.msh	$0.05 (\sin \pi y \sin \pi z, z(1 - z), 0)$; isochoric neo-Hookean, $\mu = 40$, $\kappa = 400$, mixed formulation

The mixed field has a strictly upper-triangular gradient, so $J = 1$ exactly and the exact pressure $\kappa(J - 1)$ vanishes; it is applied in four increments because on fine meshes the exact boundary data applied at once inverts the elements next to the boundary while the interior is still at rest.

Check. L^2 displacement error under `serial_refine` 1, 2, 3 (two dimensions, `test_loading`) or 1, 2 (three dimensions, `test_verification`); the rate $\log_2(e_h/e_{h/2})$ must be at least 2.8, i.e. third order for $k = 2$.

4.4 Rivlin's inflation of a thick-walled cylinder

Problem. Quarter annulus $A = 1$, $B = 2$ (`annulus.msh`, second-order geometry), plane strain, incompressible neo-Hookean with $\mu = 1$, mixed formulation; rollers on the two symmetry lines through component-wise Dirichlet data; follower pressure $P = 0.3$ per unit current area on the inner surface, ramped over the load path.

Reference (Rivlin; Ogden, *Non-linear Elastic Deformations*, Section 5.3.1). With a, b the deformed radii and $b^2 = B^2 - A^2 + a^2$,

$$P = \mu \left[\ln \frac{B}{A} - \ln \frac{b}{a} + \frac{A^2 - a^2}{2} \left(\frac{1}{b^2} - \frac{1}{a^2} \right) \right]; \quad P = 0.3 \Rightarrow a = 1.28158, b = 2.15463. \quad (1)$$

Check (`test_loading`). On the base mesh and one refinement: inner and outer radii within 0.2% of the inflation $(a - A)$, $(b - B)$; the radial displacement is the same at $(A, 0)$ and $(0, A)$ to 10^{-6} ; the rollers hold the symmetry lines to 10^{-12} ; the final Newton solve takes at most six iterations (quadratic convergence with the follower tangent); and the error decreases under refinement. The README quotes the agreement as 10^{-6} on the refined mesh.

4.5 Green–Zerna inflation of a thick-walled sphere

Problem. Octant of a spherical shell $A = 10$, $B = 11$ (`sphere_octant.msh`, curved second-order tetrahedra), incompressible neo-Hookean $\mu = 1$, mixed formulation, rollers on the three symmetry planes, follower pressure $P = 0.100975807337556$ on the inner surface.

Reference (Green and Zerna; Ogden). With $\lambda_a = a/A$, $\lambda_b = b/B$ and $b^3 = B^3 - A^3 + a^3$,

$$P = \frac{\mu}{2} \left[\left(\frac{4}{\lambda_b} + \frac{1}{\lambda_b^4} \right) - \left(\frac{4}{\lambda_a} + \frac{1}{\lambda_a^4} \right) \right], \quad (2)$$

and the chosen P gives $\lambda_a = 1.2$ exactly ($a = 12$, $b = 12.72190$), below the pressure maximum of this shell near $\lambda_a = 1.4$. The test inverts the relation by bisection.

Check (`test_verification`). Inner and outer radii within 0.5% of the inflation; the inner radius on the z axis equals that on the x axis to $10^{-3}(a - A)$ (spherical symmetry).

4.6 Rivlin's torsion of a solid cylinder

Problem. Solid cylinder $a = 12.7$, $L = 25.4$ (`cylinder_torsion.msh`, curved second-order tetrahedra), incompressible neo-Hookean $\mu = 1$, mixed formulation. The bottom face is clamped and the top face rotated rigidly by $\theta = 1$ rad with $u_z = 0$ ($\mathbf{u} = \mathbf{R}(\psi z)\mathbf{X} - \mathbf{X}$ on the face, twist $\psi = \theta/L$, rim shear $\psi a = 0.5$) in 20 increments; a larger first increment leaves the elements under the rotated face inverted before Newton starts.

Reference. Rivlin's universal solution holds exactly: $\mathbf{u} = \mathbf{R}(\psi z)\mathbf{X} - \mathbf{X}$ throughout, the lateral surface is traction free by itself, and in cylindrical components with $\gamma = \psi r$

$$\sigma_{rz} = 0, \quad \sigma_{\theta z} = \mu\psi r, \quad \sigma_{rr} = \sigma_{zz} = -\frac{1}{2}\mu\psi^2(a^2 - r^2), \quad \sigma_{\theta\theta} = \sigma_{rr} + \mu\psi^2 r^2, \quad (3)$$

$$p = \sigma_{rr} + \frac{1}{3}\mu\psi^2 r^2, \quad \sigma_{vm} = \mu\sqrt{\gamma^4 + 3\gamma^2}. \quad (4)$$

Check (`test_verification`). Displacement field to 2×10^{-3} ; on the mid-plane the shear stress $\sigma_{\theta z}$ and the von Mises stress to 2%, the normal stresses and the pressure to 3% of the stress scale, and the absence of $r\theta$ and rz shear to 2% (curved P2 tetrahedra).

4.7 Euler–Bernoulli cantilever in the linear limit

Problem. A $10 \times 1 \times 1$ beam (`beam.msh`, $20 \times 2 \times 2$ hexahedra, $k = 2$) clamped at $x = 0$ with a downward shear traction on $x = L$ small enough that $\max |\text{Grad } \mathbf{u}| \approx 5 \times 10^{-5}$; compressible neo-Hookean, and the same problem with Saint Venant–Kirchhoff. The systems AMG options are selected in the input because the elasticity options stall on this slender mesh; the test also checks that the automatic fallback converges.

Reference. Euler–Bernoulli tip deflection $PL^3/(3EI) = 3.2 \times 10^{-4}$.

Check (`test_benchmarks`). $\max |\text{Grad } \mathbf{u}| \leq 10^{-4}$ (the load is in the linear regime); tip deflection within 0.15% of the beam value; the two materials agree in the linear limit.

4.8 Euler buckling of a clamped column (Southwell plot)

Problem. A $1 \times 1 \times 20$ column (`column_euler.msh`, $2 \times 2 \times 40$ hexahedra) carrying the first-mode imperfection $x, y \leftarrow 0.0025(1 - \cos 2\pi z/20)$; compressible neo-Hookean $E = 100$, $\nu = 0.3$. The base is clamped, the top held in x, y and shortened by $u_z = -0.12t$, an axial strain up to 0.6%, about 73% of the Euler strain of the clamped-clamped column, $\varepsilon_{cr} = 4\pi^2 I/(AL^2) = 8.22 \times 10^{-3}$.

Reference. Below the limit point the mid-height lateral deflection of an imperfect column grows as $u = \delta_0(P/P_{cr})/(1 - P/P_{cr})$ (Southwell), so $1/u$ is linear in $1/\varepsilon$ with slope $\varepsilon_{cr}/\delta_0$ and intercept $-1/\delta_0$.

Check (`test_verification`). A least-squares fit of $1/u$ against $1/\varepsilon$ over the load steps recovers ε_{cr} within 5% of the Euler value and the imperfection amplitude δ_0 within 15% of the geometric one (0.005).

4.9 Kirsch’s plate with a circular hole

Problem. Quarter of a 40×40 plate with a hole of radius $a = 1$ (`plate_hole_2d.msh`, quadrilaterals of size 0.04 at the hole, curved second-order geometry), plane stress, compressible neo-Hookean $E = 1000$, $\nu = 0.3$; rollers on the symmetry lines, far-field traction $\sigma_0 = 0.1$ on $x = 20$ (strain 10^{-4} , linear regime).

Reference. Kirsch’s infinite-plate solution: $\sigma_{xx} = 3\sigma_0$ at the hole top $(0, a)$, $\sigma_{yy} = -\sigma_0$ at the hole side $(a, 0)$, and along the y axis $\sigma_{xx}/\sigma_0 = 1 + a^2/(2y^2) + 3a^4/(2y^4)$. The finite width $W = 20a$ raises the concentration by a few tenths of a percent (Howland).

Check (`test_verification`). Probed Cauchy stresses σ_{xx} , σ_{yy} within 2% of $3\sigma_0$ at the probe points, and no shear stress on the symmetry axes to 1% of $3\sigma_0$.

5 Benchmarks (`apps/input/finite_elasticity/cooks_membrane/`)

Problem. Cook’s membrane: the quadrilateral with corners $(0, 0)$, $(48, 44)$, $(48, 60)$, $(0, 44)$ (`cook.msh`), plane strain, left edge clamped, uniform upward shear traction on the right edge, $k = 2$. Three inputs:

Input	Formulation	Material	Traction (resultant)
<code>cook</code>	displacement	neo-Hookean $E = 250$, $\nu = 0.3$	3.75 (60)
<code>cook_incompressible</code>	mixed Q2-Q1	isochoric neo-Hookean $\mu = 80.194$, $\kappa = \infty$	6.25 (100)
<code>cook_nearly_incompressible</code>	mixed Q2-Q1	the same with $\nu = 0.4999$ ($\kappa = 4 \times 10^5$)	6.25 (100)

Reference. There is no closed form; the benchmark is a literature standard for locking and mesh convergence. The test performs a self-convergence study (the base 4×4 mesh and four uniform refinements up to 64×64 for the compressible case, up to 32×32 for the incompressible one) and holds the finest corner displacement as a frozen regression value.

Check (`test_benchmarks`). Monotone self-convergence of the vertical displacement of the top-right corner $(48, 60)$; the frozen values $u_y = 4.905891700497$ (compressible, 64×64 ; 30.7% of the 16 mm edge) and $u_y = 6.930412595013$ (incompressible, 32×32 ; 43.3%) reproduced to 10^{-8} relative, serially and with four ranks. The final step of `make test`, `-cook-ratio-gate`, asserts the plan’s requirement that successive differences shrink by at least three per refinement. That threshold is *not* met (measured ratios 2.34, 2.45, 2.31): uniform refinement is limited by the singularity at the 108-degree clamped-free corner, so the point value converges at about $h^{1.3}$. Everything before that step is green; the threshold is kept as written rather than relaxed, so `make test` ends with this known failure.

6 Small-strain cases (`apps/input/linear_elasticity/`)

The inputs of this directory use `model: linear_elastic` and are checked by `tests/test_linear_verification` (about 30 s). Every displacement solve takes one Newton iteration and every mixed one at most two; `newton.rtol` is 10^{-8} because it only accepts the linear solve (theory manual,

Section on the solution of a linear problem), and the accuracy is set by `linear.rtol` = 10^{-13} . Several cases are the small-strain counterparts of a finite-strain input and are compared with it: the hyperelastic “linear limit” runs have to keep $|\text{Grad } \mathbf{u}|$ near 10^{-4} and their Newton tolerance above the round-off floor of their stress evaluation, whereas the amplitude of a linear run is free.

6.1 Lamé’s thick-walled cylinder

Problem. Quarter annulus $a = 1$, $b = 2$ (`annulus.msh`, second-order arcs, 16×32 quadrilaterals after two refinements, $k = 2$), plane strain, rollers on the symmetry lines, dead pressure $P = 10$ on the inner surface; $E = 1000$ with $\nu = 0.3$ in the displacement formulation (`lame_cylinder`) and $\nu = \frac{1}{2}$ in the mixed one (`lame_cylinder_incompressible`).

Reference. With $A = Pa^2/(b^2 - a^2)$ [7], $u_r = \frac{1+\nu}{E}A[(1 - 2\nu)r + b^2/r]$, $\sigma_{rr} = A(1 - b^2/r^2)$, $\sigma_{\theta\theta} = A(1 + b^2/r^2)$, $\sigma_{zz} = 2\nu A$. For $\nu = \frac{1}{2}$ the mean stress $\frac{1}{2}(\sigma_{rr} + \sigma_{\theta\theta}) = A$ is uniform, so the pressure unknown must be the constant $A = 10/3$.

Check (`test_linear_verification`). $u_r(a)$, $u_r(b)$ and u_r in the wall to 10^{-5} (measured 3×10^{-6} , the floor of the second-order arcs, which refinement keeps). The pointwise stresses and strains of the projected fields at $r = 1.25, 1.5, 1.75$ on the diagonal converge with h^2 : error ratios 3.90, 4.04, 4.02 over three refinements, 0.26% of P at the input’s level. Incompressible: u_r to 4×10^{-6} , the pressure unknown equal to A to 3×10^{-6} in the wall and 10^{-4} in the corner elements, $\sigma_{zz} = A$, and $\text{tr } \boldsymbol{\varepsilon}$ small against the strains (the constraint holds weakly, so pointwise it is $O(h^2)$, not zero). *Locking record* at $\nu = 0.4999$ on the same mesh, error of $u_r(a)$: displacement formulation $k = 1$: 50%; $k = 2$: 2.8×10^{-4} ; mixed Q2–Q1: 4×10^{-6} .

6.2 Lamé’s thick-walled sphere

Problem. Octant $a = 10$, $b = 11$ (`sphere_octant.msh`, curved second-order tetrahedra), rollers on the three symmetry planes, dead pressure $P = 1$ inside, $E = 1000$, $\nu = 0.3$ (`lame_sphere`).

Reference. With $A = Pa^3/(b^3 - a^3)$, $u_r = \frac{A}{E}[(1 - 2\nu)r + (1 + \nu)b^3/(2r^2)]$, $\sigma_{rr} = A(1 - b^3/r^3)$, $\sigma_{\theta\theta} = A(1 + b^3/(2r^3))$.

Check. $u_r(a)$, $u_r(b)$ and $u_r(a)$ on the z axis to 3×10^{-4} (measured 7×10^{-5}); the stresses at $r = 10.5$ on the diagonal to 1% of the hoop stress, no shear on the radial direction.

6.3 Kirsch’s plate, the cantilever

Problem. `kirsch_plate_with_hole` and `euler_bernoulli_cantilever3d`: the finite-strain inputs of the same names with `linear_elastic`, the plate under $\sigma_0 = 1$ so that the probed stresses are the concentration factors, the beam under a thousand times the load.

Check. Kirsch’s stresses to 2% of $3\sigma_0$ (the finite width dominates), no shear on the axes, $\sigma_{zz} = 0$ to 10^{-10} (plane stress), and agreement with the neo-Hookean run at a strain of 10^{-4} to 5×10^{-4} (measured 9×10^{-5}). Cantilever: tip deflection within 0.2% of $PL^3/(3EI)$ (measured 0.15%); compliance equal to that of the small-load neo-Hookean run to 10^{-7} (measured 10^{-9} : the transverse deflection is odd in the load, so the nonlinear correction is second order in $|\text{Grad } \mathbf{u}| = 5 \times 10^{-5}$); clamp force and moment PL with reference arms to 10^{-9} .

6.4 Manufactured solutions

Problem. The displacement fields of the finite-strain manufactured inputs on the unit square and cube (`mms_2d_plane_strain`, `mms_3d_hex`, `mms_3d_tet`), prescribed on the whole boundary, with the body force $\rho_R \mathbf{b} = -[(\lambda + \mu) \text{grad div } \mathbf{u} + \mu \Delta \mathbf{u}]$ generated by `sympy` and checked against that identity.

Check. L^2 rates for $k = 2$: 3.00 (2D, four levels), 3.01 (hexahedra, three levels), 2.93 (tetrahedra); gates 2.9, 2.9, 2.8.

6.5 Two materials by regions

Problem. `spherical_inclusion`: the octant of a cube with a centred spherical inclusion (`inclusion.msh`), matrix $E = 100$, inclusion ten times stiffer through `material.regions`, uniaxial strain of 1% with rollers on every face.

Reference. The apparent constrained modulus $M = F/(L^2 \varepsilon) = 2\Psi/(V\varepsilon^2)$ lies between the Reuss and Voigt averages of $\lambda + 2\mu$ with the volume fraction $f = \pi a^3/(6L^3)$ [9]: $143.04 < M < 213.91$. Both bounds hold for the discrete solution (affine fields are in the space; the displacement method overestimates the energy).

Check. $M = 148.86$ from the reaction and from the energy (equal to 10^{-9}), within the bounds; with the base's parameters in the region the solution is the homogeneous uniaxial-strain field and $M = \lambda + 2\mu$ to 10^{-10} .

6.6 Cook's membrane in the small-strain theory

Problem. The membrane as it was posed [11]: plane stress, $E = 1$, $\nu = \frac{1}{3}$, unit shear load, $k = 2$ (`cook_linear`); and the incompressible plane-strain variant in the mixed formulation (`cook_linear_incompressible`, $\nu = \frac{1}{2}$, Q2-Q1).

Reference. Literature values of the vertical tip displacement lie near 23.9 (23.91 to 23.96). Probing both candidate points shows that they refer to the mid-point (48, 52) of the free edge: this code gives 23.9650 there and 25.1640 at the corner (48, 60) on the 64×64 mesh.

Check. The mid-edge value within 1% of 23.96 (measured 2×10^{-4}); both values frozen to 10^{-8} ; successive-difference ratios ≥ 2 (measured 2.08–2.27 at the corner, 2.11–3.57 at mid-edge, limited by the clamped corner as for the finite-strain membrane); clamp force -1 and moment -48 with reference arms although the corner moves by $u_x = -19$ in these units. Incompressible: corner 19.4176 and mid-edge 18.4974 on 32×32 , frozen, ratios 2.9 and 3.3; the displacement formulation at $\nu = 0.4999$, $k = 2$ gives 19.164 (1.3%). Both membranes are part of `test_parallel`.

7 Dynamic cases (apps/input/dynamics/)

The inputs of this directory carry a `dynamics` block and are checked twice: by `tests/test_dynamic_verification` (54 checks, about two minutes, driving the library), and by `apps/dynamics_compare.py`, which runs the executable, reads the time histories from its per-step lines, draws them over the closed forms (`make dynamics`) and, with `-check` in `make test`, asserts the same measures

on the executable's own output. The same test compares runs with two and four ranks against the serial one (130 steps of the two bar inputs to 5×10^{-14} , 20 steps of the tube to 2×10^{-12} in the pressure). Bar and beam are the $10 \times 1 \times 1$ body of `box.geo`: `beam.msh` ($20 \times 2 \times 2$ hexahedra) and the new `bar.msh` ($100 \times 1 \times 1$), $E = 250$, $\rho_R = 2.5$, $k = 2$.

7.1 Free axial vibration of a bar

Problem (`bar_free_vibration.yaml`). Fixed-free bar, `linear_elastic` with $\nu = 0$, so that the clamp leaves the motion one-dimensional; released from rest in its first mode, $\mathbf{u}_0 = A \sin(\pi X/2L) \mathbf{e}_x$; trapezoidal rule, 2000 steps over five periods.

Reference. $u = A \sin(\pi X/2L) \cos(\omega_1 t)$ with $\omega_1 = \pi c/2L$, $c = \sqrt{E/\rho_R} = 10$; the trapezoidal rule has the numerical frequency $\tan(\tilde{\omega}\Delta t/2) = \omega\Delta t/2$, a period elongation of $(\omega\Delta t)^2/12$ to leading order.

Check. Tip history within $6.1 \times 10^{-4}A$ of the closed form over five periods (gate $10^{-3}A$; the difference is the accumulated phase error of the scheme), energy to 7×10^{-14} . At $\Delta t = 0.2$, twenty steps per period, the period read from the zero crossings grows by 8.135×10^{-3} against 8.225×10^{-3} to leading order and 8.171×10^{-3} from the dispersion relation (gates 10 and 2 percent).

7.2 Step load on the bar: d'Alembert's wave

Problem (`bar_step_load.yaml`, `bar.msh`). The same bar with an axial traction p switched on at $t = 0$ on its free end and held: no `schedule` key, since the default of a dynamic analysis is the constant schedule, on from the first instant. One time step is the time a wave needs from one node to the next; two periods.

Reference. A front of height p leaves the loaded end, doubles at the wall at $t = L/c$ and runs back and forth with the period $4L/c$: the tip displacement is a triangle wave between 0 and $2pL/E$, twice the static value, and the wall carries $2pA$ for $L/c < t < 3L/c$ and nothing for the rest of the period.

Check. With generalized- α ($\rho_\infty = 0.8$, the input) and with the trapezoidal rule: tip peak 1.9895 and 1.9897 of the static deflection, mean 1.0001; front at mid-span at $t = 0.5001$ (a line through the rise; gate one element crossing time, 0.01); wall reaction 1.0000 pA on average. The solution is discontinuous and the finite element one rings behind every front. The first overshoot of the wall reaction is made by the mesh and is the same for both schemes, 28 percent; what the dissipation removes is the ringing that follows: late in the plateau its root mean square is 0.18 percent of $2pA$ against 1.62 with the trapezoidal rule, which keeps it for ever (checked: less than half).

7.3 First bending frequency of a cantilever

Problem (`cantilever_vibration.yaml`). The beam of the small-strain cantilever case, $\nu = 0.3$, under a half-sine pulse of the tip shear traction half a period long, which feeds the first mode and little else, then free for three periods; $\rho_\infty = 0.9$, 124 steps per period.

Reference. Euler-Bernoulli, $\omega_1 = 1.875104^2 \sqrt{EI/(\rho_R AL^4)} = 0.101499$.

Check. $\omega_1 = 0.101321$ from the zero crossings after the pulse, 0.17 percent below the reference (gate 2 percent, and below it: shear deformation and rotary inertia lower the frequency of a beam of slenderness 10).

7.4 Manufactured solutions in space and time

Problem (`mms_dynamic_2d.yaml`, `mms_dynamic_3d.yaml`). $\mathbf{u}(\mathbf{X}, t) = \sin(\omega t) \mathbf{U}(\mathbf{X})$ with the fields $\mathbf{U}$ of the small-strain manufactured solutions, prescribed on the whole boundary (a moving support on every side), $\mathbf{u}_0 = \mathbf{0}$, $\mathbf{v}_0 = \omega \mathbf{U}$; the body force per unit mass is $\sin(\omega t)(\mathbf{b}_s - \omega^2 \mathbf{U})$ with $\mathbf{b}_s$ the static one, all as expressions of x, y, z, t .

Check. At a small fixed step, L^2 rates 3.006, 3.002 (2D, refinements 1 to 3) and 3.012, 3.022 (3D, 0 to 2). The order in Δt by self-convergence, in which the spatial error cancels, on the unrefined mesh with 800 to 6400 steps: ratios 4.033, 4.010 (on the refined mesh with fewer steps they are 3.74–3.78, for the reason given in Section 3.6).

7.5 Large-amplitude vibration of a neo-Hookean block

Problem (`neo_hookean_block_vibration.yaml`). Unit square, plane strain, clamped on the left, `neo_hookean`, thrown upwards with $\mathbf{v}_0 = (0, 6X)$: the free corner swings through 0.7 of the block's size with a period of about one time unit. There is no closed form; what is checked is the order of the time integration and what each scheme does with the energy.

Check. Corner displacement at $t = 0.16$ for Δt halved from 5×10^{-4} : ratios 3.921, 3.981. With $\rho_\infty = 0.8$ the energy never exceeds its initial value and ends below it (0.99964 after a cycle and a half at the input's step). With the trapezoidal rule the energy is not conserved exactly, the problem being nonlinear, but its error falls by 4.005 and 4.001 per halving.

7.6 Radial oscillation of an incompressible tube (mixed u - p)

Problem (`knowles_tube_oscillation.yaml`). Plane strain, the quarter annulus $A = 1 \leq R \leq B = 2$ of the Lamé and Rivlin cases with rollers on the symmetry lines and both surfaces free of traction, incompressible `iso_neo_hookean`, $\mu = 1$, $\rho_R = 1$, **formulation:** `mixed`, $\rho_\infty = 0.8$. The tube starts undeformed with the isochoric radial velocity $\mathbf{v}_0 = k\mathbf{X}/|\mathbf{X}|^2$, $k = 0.4$; with free surfaces no follower load is needed.

Reference (Knowles [10]). Incompressibility leaves the motion one degree of freedom, $r^2 = R^2 + c(t)$, so that $\dot{r} = \dot{c}/2r$ and $\ddot{r} = \ddot{c}/2r - \dot{c}^2/4r^3$. The spatial radial balance is $\rho_R \ddot{r} = \partial_r \sigma_{rr} + (\sigma_{rr} - \sigma_{\theta\theta})/r$, and for the neo-Hookean solid $\boldsymbol{\sigma} = -q\mathbf{I} + \mu\mathbf{B}$ with $\lambda_\theta = r/R = \lambda$, $\lambda_r = 1/\lambda$ gives $\sigma_{\theta\theta} - \sigma_{rr} = \mu(\lambda^2 - \lambda^{-2})$. Integrating from the inner radius a , where $\sigma_{rr} = 0$, with $\int \mu(\lambda^2 - \lambda^{-2}) dr/r = \frac{\mu}{2}[\ln((s-c)/s) - c/s]$, $s = r^2$:

$$\begin{aligned} \sigma_{rr}(r) = \rho_R \left[\frac{\ddot{c}}{2} \ln \frac{r}{a} + \frac{\dot{c}^2}{8} \left(\frac{1}{r^2} - \frac{1}{a^2} \right) \right] \\ + \frac{\mu}{2} \left[\ln \frac{(r^2 - c)a^2}{r^2 A^2} - \frac{c}{r^2} + \frac{c}{a^2} \right], \quad a^2 = A^2 + c. \end{aligned} \quad (5)$$

The outer surface is free as well, $\sigma_{rr}(b) = 0$ with $b^2 = B^2 + c$, which is Knowles' equation for $c(t)$,

$$\rho_R \left[\frac{\ddot{c}}{2} \ln \frac{b}{a} - \frac{\dot{c}^2}{8} \left(\frac{1}{a^2} - \frac{1}{b^2} \right) \right] = -\frac{\mu}{2} \left[\ln \frac{B^2 a^2}{A^2 b^2} + c \left(\frac{1}{a^2} - \frac{1}{b^2} \right) \right], \quad c(0) = 0, \quad \dot{c}(0) = 2k. \quad (6)$$

Small oscillations have $\omega^2 = 2\mu(A^{-2} - B^{-2})/(\rho_R \ln(B/A))$, a period of 4.27; at this amplitude the inner radius swings between 0.91 and 1.28 with the period 4.2428. The pressure unknown of the mixed formulation is $p = \sigma_{rr} - \mu \operatorname{dev}(\mathbf{B})_{rr}$ with $\operatorname{dev}(\mathbf{B})_{rr} = (2R^2/r^2 - r^2/R^2 - 1)/3$ (the plane-strain $\mathbf{B}$ has $B_{zz} = 1$), so (5) gives it at every material radius. The test integrates the equation with a Runge–Kutta scheme of its own (40 substeps per time step), the script with `scipy`.

Check. Inner displacement: first peak 0.280754 against 0.280749, largest difference over two periods 2.3×10^{-3} of it (gate 5×10^{-3} ; the outer wall 6×10^{-4}), period 4.2431 against 4.2428. Pressure at mid-wall, $R = 1.5$, where it is carried by the inertia of the material inside it (at the wall the free surface ties it to the kinematics): within 2.0×10^{-3} of its largest value 0.3385 after the start-up (gate 2 percent; $t > 0.5$). The start-up is visible in the pressure of the first steps: the initial pressure of an incompressible material is taken as zero and the error returns with the factor $-\rho_\infty$ (Section 3.6).

The linear solver. This case is also the measurement behind the inertial part of the Schur complement approximation (theory manual, saddle-point solver). Outer FGMRES iterations of the first Newton solve; the quasi-static inflation on the same mesh needs 18:

Δt	0.125	0.02	0.0125	0.002
refine 1, pressure mass alone	28	33	38	43
refine 2, pressure mass alone	33	47	59	81
refine 1, with the inertial part	14	8	8	6
refine 2, with the inertial part	16	9	7	6

8 Cross-check with an independent code: the elastic bar (apps/input/elastic_bar/)

Problem. The exercise of `myapps/elastic_bar`: a $0.3 \times 0.3 \times 1$ bar (`elastic_bar.msh`, the exercise’s own mesh file: 954 nodes, linear tetrahedra), clamped on $z = 0$, the face $z = 1$ pulled to $u_z = 3$ (a stretch of 4) with free lateral motion, $\mu = 5 \times 10^6$, $\kappa = 1.5 \times 10^9$ ($\nu = 0.4983$); the reaction force on the pulled face is recorded against its displacement over 100 load steps. The exercise has two drivers, a bilinear form with MFEM’s `ElasticityIntegrator` and a nonlinear form with `HyperelasticNLFIntegrator`, a hand-coded compressible Gent model ($J_m = 50$) and PETSc solvers. Here the two bars are inputs of the general executable that differ in the material line: `bar_linear` (`linear_elastic`), `bar_gent` (`gent_compressible_summit`, the same energy), and `bar_neo_hookean` (`iso_neo_hookean`, the exercise’s alternative model). `output.reactions` and a probe on the pulled face print force and displacement after every step; `apps/elastic_bar_compare.py` runs the inputs, writes the exercise’s `force_displacement.csv` format, tabulates and plots.

Reference. The curves of the exercise’s own drivers, kept in `apps/input/elastic_bar/reference/`: the stored 500-step curve of the nonlinear driver, and the linear driver run on the same mesh with its Krylov tolerance tightened from 10^{-8} to 10^{-14} . Mesh, order and boundary conditions are identical, so the discrete problems are the same and the two codes must agree to solver tolerance. In addition, the homogeneous uniaxial-stress estimate $F = A P_{11}(\lambda_1)$ of each model with the lateral stretch from $P_{22} = 0$.

Check (`apps/elastic_bar_compare.py -check`, part of `make test`; `make elastic_bar` for the plot). Largest relative difference of the force over the whole curve: 9.9×10^{-13} for the Gent bar

and 2.6×10^{-11} for the linear bar (gate 10^{-8}); against the linear driver with its own tolerance of 10^{-8} the difference is 1.6×10^{-5} , which is that tolerance. Measured forces at $u_z = 3$: 4.5697×10^6 (linear), 2.6332×10^6 (Gent), 1.9138×10^6 (neo-Hookean). The Gent curve starts with the slope 0.905×10^6 against 1.523×10^6 of the linear bar with the same μ and κ , because κ scales a quartic penalty there and does not enter the small-strain response ($\lambda = 2\mu/J_m$); the neo-Hookean curve leaves the origin along the linear one. The linear bar is 13.0% stiffer than homogeneous uniaxial stress EA/H ; with $k = 2$ it is 3.3%, so ten of the thirteen points are the volumetric locking of linear tetrahedra at this Poisson ratio and three the clamped end.

9 Kirsch's plate with exact boundary data (plate_with_hole/)

Problem. The exercise of `myapps/plate_with_hole`: a quarter plate $[0, 3R]^2$ with a quarter hole of radius $R = 0.01$ at the origin (`plate_with_a_hole.msh`, the exercise's own mesh: 10900 nodes, straight-sided triangles), plane strain, `linear_elastic` with $\mu = 7 \times 10^{10}$, $\nu = 0.3$. The displacement of the infinite plate under the far-field stress $\sigma_\infty = 0.01\mu$ is prescribed on the four outer edges and the hole is traction free. The exercise is a driver of its own; here it is the input `plate_linear` of the general executable, with the Dirichlet data written as expressions in x and y .

Reference. Kirsch's solution [4, 7] with $k = 3 - 4\nu$,

$$u_r = \frac{\sigma_\infty}{8\mu r} \left[(k-1)r^2 + 2R^2 + 2(R^2(k+1) + r^2 - R^4/r^2) \cos 2\theta \right],$$

$$u_\theta = -\frac{\sigma_\infty}{4\mu r} \left[r^2 + R^2(k-1) + R^4/r^2 \right] \sin 2\theta,$$

and the stresses of Section 6. Because the exact field is imposed on the outer boundary there is no finite-width error (the case `kirsch_plate_with_hole` of Section 6 pulls a finite plate by a traction and carries one): the finite element solution converges to the closed form, up to the geometry of the mesh, whose hole is a polygon with a chord sag of $10^{-4}R$ that uniform refinement keeps.

Check (`test_linear_verification`; `apps/plate_with_hole_compare.py -check` in `make test`; `make plate_with_hole` for the plot). The expressions of the input equal an independent implementation of the closed form to 10^{-13} . Relative L^2 error of the displacement 5.3×10^{-5} for $k = 1$ (gate 10^{-4}) and 1.4×10^{-5} for $k = 2$ (gate 3×10^{-5} , the geometry floor); σ_{xx} just inside the top of the hole and σ_{yy} at its side within 1% and 0.3% of $3\sigma_\infty$; $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$. The script reads the ParaView output back and compares every node: root-mean-square errors 5.5×10^{-5} (displacement) and $2.6 \times 10^{-3}\sigma_\infty$ (σ_{xx}) at $k = 1$, 9.1×10^{-4} after one refinement and 2.1×10^{-4} at $k = 2$; concentration factor 2.991, 2.997, 2.998. With `neo_hookean` and the same moduli the displacement differs from Kirsch's by 2.9×10^{-4} : at these strains geometric nonlinearity is five times the discretisation error.

10 Validation against a reference code: Anand's examples

The inputs of `apps/input/anand_coupled_theories/finite_elasticity/` reproduce a published set of finite element examples.

Reference. The ten "Finite Elasticity" examples of Lallit Anand's *Introduction to Coupled Theories in Solid Mechanics* (Oxford University Press, 2025), from the FEniCSx companion codes by Eric Stewart and Lallit Anand ([solidmechanicscoupletheories.github.io](https://github.com/solidmechanicscoupletheories)). All use Arruda–Boyce with $G_0 = 280$ kPa, $\lambda_L = 5.12$ ($N = \lambda_L^2 = 26.2144$) and $K = 1000 G_0$, in kPa and mm, the reference's logarithmic volumetric law $p = K \ln J/J$ (volumetric: `logarithmic`, run with `solver.predictor: tangent`), the mixed formulation on Q2–Q1 or P2–P1 elements, and the reference's geometry, boundary conditions, load histories and step counts. Every input probes the points of the reference's plots after every step and prints the reactions, so the curves can be read from the log. These are validation cases: the comparison is with another code's results and with analytical curves where they exist, not a test with a tolerance.

Input	Reference	Problem
01_uniaxial_tension	3D01	10 mm cube, rollers on three planes, stretch 7.75 in y over 100 steps
02_simple_shear	3D02	1 mm cube, base clamped, top face $u_x = \sin 4\pi t$ (two cycles of shear strain 1)
03_cylinder_torsion	3D03	cylinder $R = 12.7$, $L = 25.4$, top face rotated rigidly by 2.5 rad with $u_z = 0$, 20 steps
04_plate_with_hole	3D04	quarter plate $15 \times 10 \times 1$ with a 3 mm hole, stretch 3 in x
05_cylinder_inflation	3D05	quarter tube $10/11 \times 5$, plane-strain rollers, follower pressure to 50 kPa
06_sphere_inflation	3D06	octant shell $10/11$, follower pressure to 35 kPa
07_cube_footing	3D07	50 mm cube, follower pressure 1500 kPa on a quarter patch of the top, held in x, y there
08_column_buckling	3D08	$1 \times 1 \times 20$ column with the first-mode imperfection, shortened by 2.5 mm
09_spherical_inclusion	3D09	octant of a cube with a ten times stiffer spherical inclusion (<code>material.regions</code>), stretch 2
10_column_twist	3D10	$1 \times 1 \times 3$ column, top face turned through 2π

Differences from the reference that change the numbers. The Arruda–Boyce model here is the five-term series in I_1/N rather than the Padé approximation of the inverse Langevin function, so it is softer near the locking stretch (visible in case 01 above a stretch of about 3); case 02 keeps the quadratic law $p = K(J - 1)$, because with the logarithmic law the run fails at $t = 0.064$ for any increment size: the tangent bulk modulus $K(1 - \ln J)/J^2$ of that law softens in dilatation ($0.18 K$ at $J = 1.67$, zero at $J = e$), and at the clamped top corners, stress singularities that this mesh and quadrature resolve sharply, J grows with the shear until the volumetric response is lost; hexahedra replace tetrahedra on the boxes; and a failed increment is bisected instead of ending the run (cases 05 and 06 stop early in the reference).

Plots. `apps/anand_plots.py` reproduces the result plots of the reference pages from the logs (save the stdout of each run as `out/anand_coupled_theories/finite_elasticity/logs/<case>.log`, or pass `-logs`) and overlays an analytical curve where one exists: the homogeneous incompressible and nearly incompressible Arruda–Boyce response for the uniaxial and shear blocks (with both this code's series and the reference's Padé form), Rivlin's universal torsion for torque and axial force by quadrature in r , the incompressible thick-walled cylinder and sphere inflation by quadrature with their limit points, the Euler load $4\pi^2 EI/L^2$ for the clamped column, and the matrix-only curve for the inclusion.

What the plots show. The uniaxial cube, the torsion and both inflations lie on their reference curves (the uniaxial one on the nearly incompressible solution at $K = 1000G$, with the incompressible limit a few percent above at the largest stretch). The sheared block carries about 17% less nominal shear stress than homogeneous simple shear at a shear strain of 1, because its lateral faces are free where simple shear needs tractions, and its two cycles retrace one curve (elastic, no hysteresis). The sphere stops at its limit pressure of 34 kPa and the cylinder near 38 kPa, where the reference also stopped. The buckling column reaches 7.3 mN at 0.2 mm of shortening, 5% above the Euler load of the clamped column, and then rises slowly to 7.5 mN at 2.5 mm along the hardening post-buckling path of the elastica (the Euler value uses $E = 3G$ and neglects the finite section). The twisted column needs a compressive axial force of 78 mN to keep its length over a full turn (Poynting effect), as does the torsion cylinder.

11 Summary

Case	Reference	Check	Test
homogeneous deformations	closed forms (six models)	all probed fields, 10^{-8}	<code>homogeneous_compare.py</code>
compressible uniaxial	lateral stretch from $P_{22} = 0$	$u, P_{11}, \sigma_{11}, J$ to 10^{-7}	<code>test_verification</code>
manufactured solutions	exact field, sympy body force	L^2 rate ≥ 2.8 ($k = 2$)	<code>test_loading, test_verification</code>
cylinder inflation	Rivlin	radii to 0.2%, refinement	<code>test_loading</code>
sphere inflation	Green–Zerna	radii to 0.5%, symmetry	<code>test_verification</code>
torsion	Rivlin universal solution	u to 2×10^{-3} , stresses 2–3%	<code>test_verification</code>
cantilever	Euler–Bernoulli	tip deflection to 0.15%	<code>test_benchmarks</code>
column buckling	Euler load, Southwell	ε_{cr} to 5%	<code>test_verification</code>
plate with hole	Kirsch	stresses to 2%	<code>test_verification</code>
Cook’s membrane	frozen values, self-convergence	10^{-8} ; ratio gate not met	<code>test_benchmarks</code>
Anand examples	FEniCSx codes, analytical curves	plots	<code>anand_plots.py</code>
parallel consistency	serial run	norms to 10^{-10}	<code>test_parallel</code>
linear operator	MFEM <code>ElasticityIntegrator</code> , hyperelastic tangents at $\mathbf{u} = \mathbf{0}$	10^{-12}	<code>test_linear_elasticity</code>
Hooke states, outputs	closed forms at 5% strain	all presentations, 10^{-10}	<code>test_linear_elasticity</code>
Lamé cylinder	Lamé; uniform pressure at $\nu = \frac{1}{2}$	u_r to 10^{-5} , stresses by h^2	<code>test_linear_verification</code>
Lamé sphere	Lamé	u_r to 3×10^{-4} , stresses 1%	<code>test_linear_verification</code>
plate, cantilever (small strain)	Kirsch, Euler–Bernoulli, the finite-strain runs	2%, 0.2%; 5×10^{-4} , 10^{-7}	<code>test_linear_verification</code>
manufactured (small strain)	exact field, analytic body force	L^2 rate ≥ 2.8 ($k = 2$)	<code>test_linear_elasticity, test_linear_verification</code>
two materials	Reuss and Voigt bounds	bounds; reaction vs energy	<code>test_linear_verification</code>
Cook’s membrane (small strain)	literature 23.96, frozen values	1%; 10^{-8}	<code>test_linear_verification</code>
elastic bar (linear, Gent)	the drivers of <code>myapps/elastic_bar</code>	force curves to 10^{-8} (measured 10^{-11} , 10^{-12})	<code>elastic_bar_compare.py</code>
plate with hole, exact boundary data	Kirsch over the whole field	u to 10^{-4} ($k = 1$), 3×10^{-5} ($k = 2$); stresses 1%	<code>test_linear_verification, plate_with_hole_compare.py</code>
inertia, time integration	free fall, exact-in-space field, energy identity, parameter map	10^{-11} ; order 2.00; 10^{-10} ; decay ρ_∞^{2n}	<code>test_dynamics</code>
bar, first mode	$A \cos(\omega_1 t)$; dispersion of the trapezoidal rule	tip to $10^{-3}A$; elongation to 2%	<code>test_dynamic_verification, dynamics_compare.py</code>
bar, step load	d’Alembert	23 peak 2.00, mean 1.00 to 2%; front to 0.01; ringing	same
cantilever frequency	Euler–Bernoulli	ω_1 to 2% (measured 0.17%)	same

A Closed-form homogeneous states of incompressible models

The reference solutions of Section 4.1 and of `tests/test_homogeneous` are collected here [1]. The models themselves (Ψ , its derivatives and small-strain shear modulus for the neo-Hookean, Mooney–Rivlin, Yeoh, Gent, Arruda–Boyce and Ogden energies) are given in the theory manual.

A.1 Stress of an incompressible isotropic solid

With $J = \lambda_1 \lambda_2 \lambda_3 = 1$ the second invariant is $I_2 = \lambda_1^{-2} + \lambda_2^{-2} + \lambda_3^{-2} = \text{tr } \mathbf{B}^{-1}$ and Cayley–Hamilton gives $\mathbf{B}^2 = I_1 \mathbf{B} - I_2 \mathbf{I} + \mathbf{B}^{-1}$. For $\Psi = \Psi(I_1, I_2)$ write $\Psi_1 = \partial\Psi/\partial I_1$, $\Psi_2 = \partial\Psi/\partial I_2$. The constraint is enforced by a multiplier that does no work and is fixed by equilibrium and the boundary conditions, not by the constitutive law; only the isotropic part of the stress is affected, so the convention-free statement is the deviatoric–mean split

$$\boxed{\boldsymbol{\sigma} = \boldsymbol{\sigma}' + p \mathbf{I}, \quad \boldsymbol{\sigma}' = 2\Psi_1 \text{dev } \mathbf{B} - 2\Psi_2 \text{dev } \mathbf{B}^{-1}, \quad p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}.} \quad (7)$$

Much of the literature writes $\boldsymbol{\sigma} = -\tilde{p} \mathbf{I} + 2\Psi_1 \mathbf{B} - 2\Psi_2 \mathbf{B}^{-1}$ and calls $\tilde{p}$ the hydrostatic pressure; because authors absorb different isotropic terms into $\tilde{p}$ its value depends on the convention, whereas $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$ (tensile positive) does not, and it is the unknown of the mixed formulation and the **pressure** output. A compressive hydrostatic state has $p < 0$. In the principal frame

$$\sigma_i = 2\Psi_1 \lambda_i^2 - 2\Psi_2 \lambda_i^{-2} - q = \lambda_i \frac{\partial\Psi}{\partial \lambda_i} - q', \quad \sigma_i - \sigma_j = 2\Psi_1 (\lambda_i^2 - \lambda_j^2) - 2\Psi_2 (\lambda_i^{-2} - \lambda_j^{-2}) = \lambda_i \frac{\partial\Psi}{\partial \lambda_i} - \lambda_j \frac{\partial\Psi}{\partial \lambda_j}, \quad (8)$$

so the principal stress differences are constitutive, and the recipe for every homogeneous problem is the same: the two differences follow from (8), one principal stress is fixed by a traction-free face, and then $p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$. With $J = 1$ the Piola–Kirchhoff stresses are $\mathbf{P} = \boldsymbol{\sigma} \mathbf{F}^{-T}$ and $\mathbf{S} = \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-T}$; for $\mathbf{F} = \text{diag}(\lambda_i)$, $P_i = \sigma_i / \lambda_i$ (the nominal stress that a traction condition of the total Lagrangian form prescribes) and $S_i = \sigma_i / \lambda_i^2$. For small strains $\boldsymbol{\sigma} \approx 2\mu \boldsymbol{\varepsilon} + p \mathbf{I}$ with $\mu = 2(\Psi_1 + \Psi_2)|_{I_1=I_2=3}$, $E = 3\mu$, $\nu = \frac{1}{2}$.

Models in I_1 alone. Yeoh, Gent and Arruda–Boyce have $\Psi_2 = 0$, so (7) reduces to the neo-Hookean form with a deformation-dependent modulus, $\boldsymbol{\sigma} = \mu_{\text{eff}}(I_1) \text{dev } \mathbf{B} + p \mathbf{I}$, $\mu_{\text{eff}} = 2\Psi_1(I_1)$: every neo-Hookean result below applies to them with μ replaced by μ_{eff} at the I_1 of that deformation. Ogden’s model is not an invariant model and is evaluated in the principal frame with (8), $\lambda_i \partial\Psi/\partial \lambda_i = \sum_r \mu_r \lambda_i^{\alpha_r}$.

A.2 The four modes

A homogeneous deformation has constant $\mathbf{F}$, hence constant $\boldsymbol{\sigma}$ and $\mathbf{P}$; $\text{Div } \mathbf{P} = 0$ holds identically, so the state is an exact solution without body force and p is a constant fixed by the traction-free faces. Extension is $\lambda > 1$, compression $\lambda < 1$; the formulas cover both. NH is neo-Hookean, MR Mooney–Rivlin with $\mu = 2(c_1 + c_2)$.

Uniaxial tension and compression. $\mathbf{F} = \text{diag}(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$, $I_1 = \lambda^2 + 2/\lambda$, $I_2 = 2\lambda + 1/\lambda^2$, $\sigma_2 = \sigma_3 = 0$:

$$\boxed{\sigma_1 = 2\left(\lambda^2 - \frac{1}{\lambda}\right)\left(\Psi_1 + \frac{\Psi_2}{\lambda}\right), \quad P_1 = \frac{\sigma_1}{\lambda}, \quad p = \frac{\sigma_1}{3}.} \quad (9)$$

$$\text{NH: } \sigma_1 = \mu \left(\lambda^2 - \frac{1}{\lambda} \right), \quad P_1 = \mu \left(\lambda - \frac{1}{\lambda^2} \right), \quad (10)$$

$$\text{MR: } \sigma_1 = 2 \left(c_1 + \frac{c_2}{\lambda} \right) \left(\lambda^2 - \frac{1}{\lambda} \right), \quad P_1 = 2 \left(c_1 + \frac{c_2}{\lambda} \right) \left(\lambda - \frac{1}{\lambda^2} \right), \quad (11)$$

$$\text{Ogden: } \sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r/2}), \quad P_1 = \sum_r \mu_r (\lambda^{\alpha_r-1} - \lambda^{-\alpha_r/2-1}). \quad (12)$$

Small strain: $\sigma_1 \approx 3\mu(\lambda - 1) = E(\lambda - 1)$.

Equibiaxial tension. $\mathbf{F} = \text{diag}(\lambda, \lambda, \lambda^{-2})$, $I_1 = 2\lambda^2 + 1/\lambda^4$, $I_2 = \lambda^4 + 2/\lambda^2$, $\sigma_3 = 0$:

$$\boxed{\sigma_1 = \sigma_2 = 2 \left(\lambda^2 - \frac{1}{\lambda^4} \right) (\Psi_1 + \lambda^2 \Psi_2), \quad P_1 = P_2 = \frac{\sigma_1}{\lambda}, \quad p = \frac{2\sigma_1}{3}.} \quad (13)$$

NH: $\sigma_1 = \mu(\lambda^2 - \lambda^{-4})$; MR: $\sigma_1 = 2(c_1 + c_2\lambda^2)(\lambda^2 - \lambda^{-4})$; Ogden: $\sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-2\alpha_r})$. Equibiaxial tension to λ is uniaxial compression to λ^{-2} plus a hydrostatic stress, which is why the two tests are interchangeable for identifying Ψ .

Pure shear (planar tension). $\mathbf{F} = \text{diag}(\lambda, 1, \lambda^{-1})$, $I_1 = I_2 = \lambda^2 + 1 + 1/\lambda^2$, $\sigma_3 = 0$:

$$\boxed{\begin{aligned} \sigma_1 &= 2 \left(\lambda^2 - \frac{1}{\lambda^2} \right) (\Psi_1 + \Psi_2), & \sigma_2 &= 2 \left(1 - \frac{1}{\lambda^2} \right) (\Psi_1 + \lambda^2 \Psi_2), \\ P_1 &= \frac{\sigma_1}{\lambda}, & P_2 &= \sigma_2, & p &= \frac{\sigma_1 + \sigma_2}{3}. \end{aligned}} \quad (14)$$

$$\text{NH: } \sigma_1 = \mu \left(\lambda^2 - \frac{1}{\lambda^2} \right), \quad \sigma_2 = \mu \left(1 - \frac{1}{\lambda^2} \right), \quad (15)$$

$$\text{MR: } \sigma_1 = \mu \left(\lambda^2 - \frac{1}{\lambda^2} \right), \quad \sigma_2 = 2(c_1 + c_2\lambda^2) \left(1 - \frac{1}{\lambda^2} \right), \quad (16)$$

$$\text{Ogden: } \sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r}), \quad \sigma_2 = \sum_r \mu_r (1 - \lambda^{-\alpha_r}). \quad (17)$$

The principal stretches $(\lambda, 1, \lambda^{-1})$ are those of a simple shear of amount $\gamma = \lambda - \lambda^{-1}$: pure shear is simple shear followed by a rotation. For Mooney–Rivlin σ_1 depends on μ alone; the constrained-direction stress σ_2 is what distinguishes c_1 from c_2 .

Simple shear. $\mathbf{F} = \mathbf{I} + \gamma \mathbf{e}_1 \otimes \mathbf{e}_2$,

$$\mathbf{B} = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}^{-1} = \begin{pmatrix} 1 & -\gamma & 0 \\ -\gamma & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad I_1 = I_2 = 3 + \gamma^2. \quad (18)$$

The principal stretches are $\lambda = \frac{1}{2}(\gamma + \sqrt{\gamma^2 + 4})$, λ^{-1} and 1, with the in-plane principal directions at the angle θ to $\mathbf{e}_1$, $\tan 2\theta = 2/\gamma$. The deviatoric stress (7) is $\sigma'_{12} = 2(\Psi_1 + \Psi_2)\gamma$, $\sigma'_{11} = \frac{2}{3}(2\Psi_1 + \Psi_2)\gamma^2$, $\sigma'_{22} = -\frac{2}{3}(\Psi_1 + 2\Psi_2)\gamma^2$, $\sigma'_{33} = -\frac{2}{3}(\Psi_1 - \Psi_2)\gamma^2$, which gives the shear stress and the normal-stress differences independently of p :

$$\boxed{\begin{aligned} \sigma_{12} &= 2(\Psi_1 + \Psi_2)\gamma, & \sigma_{11} - \sigma_{22} &= \gamma\sigma_{12} = 2(\Psi_1 + \Psi_2)\gamma^2, \\ \sigma_{11} - \sigma_{33} &= 2\Psi_1\gamma^2, & \sigma_{22} - \sigma_{33} &= -2\Psi_2\gamma^2. \end{aligned}} \quad (19)$$

The secant shear modulus $\sigma_{12}/\gamma = 2(\Psi_1 + \Psi_2)$ is constant for NH and MR (a linear shear response) and depends on γ through $I_1 = 3 + \gamma^2$ for the models in I_1 alone; the relation $\sigma_{11} - \sigma_{22} = \gamma\sigma_{12}$ is universal. The mean stress depends on which face is traction free:

$$\sigma_{33} = 0 \text{ (thin sheet): } \sigma_{11} = 2\Psi_1\gamma^2, \quad \sigma_{22} = -2\Psi_2\gamma^2, \quad p = \frac{2}{3}(\Psi_1 - \Psi_2)\gamma^2, \quad (20)$$

$$\sigma_{22} = 0 \text{ (free top face, plane strain): } \sigma_{11} = 2(\Psi_1 + \Psi_2)\gamma^2, \quad \sigma_{33} = 2\Psi_2\gamma^2, \quad p = \frac{2}{3}(\Psi_1 + 2\Psi_2)\gamma^2. \quad (21)$$

NH (either case): $\sigma_{12} = \mu\gamma$, $\sigma_{11} = \mu\gamma^2$, $\sigma_{22} = \sigma_{33} = 0$, $p = \mu\gamma^2/3$. MR with $\sigma_{33} = 0$: $\sigma_{11} = 2c_1\gamma^2$, $\sigma_{22} = -2c_2\gamma^2$, $p = \frac{2}{3}(c_1 - c_2)\gamma^2$; with $\sigma_{22} = 0$: $\sigma_{11} = \mu\gamma^2$, $\sigma_{33} = 2c_2\gamma^2$, $p = \frac{2}{3}(c_1 + 2c_2)\gamma^2$. Ogden:

$$\sigma_{12} = \frac{\sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r})}{\lambda + \lambda^{-1}}, \quad \sigma_{11} - \sigma_{22} = \gamma\sigma_{12}, \quad \sigma_{11} + \sigma_{22} - 2\sigma_{33} = \sum_r \mu_r (\lambda^{\alpha_r} + \lambda^{-\alpha_r} - 2). \quad (22)$$

The nominal stress follows from $\mathbf{F}^{-T} = \mathbf{I} - \gamma \mathbf{e}_2 \otimes \mathbf{e}_1$: $P_{11} = \sigma_{11} - \gamma\sigma_{12}$, $P_{12} = \sigma_{12}$, $P_{21} = \sigma_{12} - \gamma\sigma_{22}$, $P_{22} = \sigma_{22}$, $P_{33} = \sigma_{33}$, so that for NH $P_{11} = P_{22} = 0$ and $P_{12} = P_{21} = \mu\gamma$: the nominal tractions on the sheared faces are purely tangential. For any other model the lateral faces $X_1 = \text{const}$ must carry the normal nominal traction P_{11} for simple shear to be an exact solution.

Consistency checks. All stresses vanish at $\lambda = 1$ or $\gamma = 0$. The initial slopes are $d\sigma_1/d\lambda = 3\mu$ (uniaxial), 6μ (equibiaxial), 4μ (pure shear) and $d\sigma_{12}/d\gamma = \mu$, and $\sigma_{11} - \sigma_{22} = \gamma\sigma_{12}$ holds in simple shear for every isotropic incompressible model.

Mode (NH)	Stretches	I_1	Non-zero Cauchy stress	$p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$
Uniaxial	$(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$	$\lambda^2 + 2\lambda^{-1}$	$\sigma_1 = \mu(\lambda^2 - \lambda^{-1})$	$\sigma_1/3$
Equibiaxial	$(\lambda, \lambda, \lambda^{-2})$	$2\lambda^2 + \lambda^{-4}$	$\sigma_1 = \sigma_2 = \mu(\lambda^2 - \lambda^{-4})$	$2\sigma_1/3$
Pure shear	$(\lambda, 1, \lambda^{-1})$	$\lambda^2 + 1 + \lambda^{-2}$	$\sigma_1 = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_2 = \mu(1 - \lambda^{-2})$	$(\sigma_1 + \sigma_2)/3$
Simple shear	$(\lambda, \lambda^{-1}, 1)$, $\gamma = \lambda - \lambda^{-1}$	$3 + \gamma^2$	$\sigma_{12} = \mu\gamma$, $\sigma_{11} = \mu\gamma^2$	$\mu\gamma^2/3$

For Yeoh, Gent and Arruda–Boyce replace μ by $2\Psi_1(I_1)$ at the tabulated I_1 .

A.3 Reproducing the states in two dimensions

In two dimensions the kernels build $\mathbf{F}$ from the in-plane block with $F_{33} = 1$ (plane strain) or the thickness stretch (plane stress). An incompressible plane-strain state has in-plane principal stretches (λ, λ^{-1}) : up to a rotation it is the pure shear (14) with $\mathbf{e}_3$ as the constrained direction, and the reaction σ_{33} enters both p and the von Mises stress.

Plane-strain extension. Stretch λ along $\mathbf{e}_1$ with the faces $X_2 = 0, H$ traction free. Relabelling (14) with $(\lambda_1, \lambda_2, \lambda_3) = (\lambda, \lambda^{-1}, 1)$,

$$\mathbf{u} = (\lambda - 1)X_1 \mathbf{e}_1 + (\lambda^{-1} - 1)X_2 \mathbf{e}_2, \quad (23)$$

$$\sigma_{11} = 2(\Psi_1 + \Psi_2) \left(\lambda^2 - \frac{1}{\lambda^2} \right), \quad \sigma_{33} = 2 \left(1 - \frac{1}{\lambda^2} \right) (\Psi_1 + \lambda^2 \Psi_2), \quad (24)$$

$$p = \frac{\sigma_{11} + \sigma_{33}}{3}, \quad P_{11} = \frac{\sigma_{11}}{\lambda}, \quad \sigma_{vm} = \sqrt{\sigma_{11}^2 + \sigma_{33}^2 - \sigma_{11}\sigma_{33}}; \quad (25)$$

NH: $\sigma_{11} = \mu(\lambda^2 - \lambda^{-2})$, $\sigma_{33} = \mu(1 - \lambda^{-2})$, $p = \frac{1}{3}\mu(\lambda^2 + 1 - 2\lambda^{-2})$; MR: the same σ_{11} with $\sigma_{33} = 2(c_1 + c_2\lambda^2)(1 - \lambda^{-2})$. In an input the affine $\mathbf{u}$ is prescribed on the two end faces and $X_2 = 0, H$ are left free (or $\mathbf{u}$ on $X_1 = 0$ and the nominal traction $P_{11}\mathbf{e}_1$ on $X_1 = L$); with Taylor–Hood spaces the affine displacement and the constant pressure lie in the discrete spaces, so the mixed formulation reproduces them to solver tolerance.

Plane-strain simple shear. $\mathbf{u} = \gamma X_2 \mathbf{e}_1$. With the displacement prescribed on the whole boundary (the patch test of `tests/test_mixed`) the incompressible pressure is determined only up to a constant (Piola identity), so the checkable quantities are the deviatoric stress and the relations (19); at finite κ the affine field is exactly isochoric, $p = \kappa(J - 1) = 0$ and $\boldsymbol{\sigma} = \boldsymbol{\sigma}'$. If only the top and bottom faces are constrained and the lateral faces are left free, simple shear is *not* the exact solution (the lateral faces would have to carry P_{11} and P_{21}); even for NH, where $P_{11} = 0$, the free faces perturb the state near the edges because $P_{21} = \mu\gamma \neq 0$.

Plane stress. For a thin sheet $\sigma_{i3} = 0$ and the thickness stretch is set by the material. For an incompressible material $\lambda_3 = 1/\det \mathbf{F}_{2D}$ and $\sigma_{33} = 0$ fixes the mean stress as $p = -\sigma'_{33}$,

$$\sigma_{\alpha\beta} = \sigma'_{\alpha\beta} - \sigma'_{33} \delta_{\alpha\beta}, \quad \mathbf{P} = \mathbf{P}_{\text{iso}} + p \mathbf{F}^{-T}, \quad p = -\lambda_3 P_{\text{iso},33}, \quad (26)$$

so the sheet is a displacement problem in the two in-plane components with no pressure unknown, no inf-sup condition and no locking. The four modes above are all plane-stress states and become two-dimensional problems: the uniaxial sheet with $\mathbf{u} = (\lambda - 1)X_1\mathbf{e}_1 + (\lambda^{-1/2} - 1)X_2\mathbf{e}_2$ on the end faces and the lateral faces free, and equibiaxial tension, pure shear and thin-sheet simple shear with the affine displacement on the whole boundary, which is well posed here because p is determined pointwise. Since each load step starts with the boundary displaced and the interior lagging, shear-dominated sheet problems need increments that shear a boundary layer of elements by no more than a few percent, as the energy grows like $(\det \mathbf{F}_{2D})^{-2}$.

References

- [1] R. W. Ogden, *Non-Linear Elastic Deformations*, Ellis Horwood, 1984 (Dover, 1997).
- [2] R. S. Rivlin, Large elastic deformations of isotropic materials IV, *Phil. Trans. R. Soc. A* 241 (1948) 379–397.
- [3] A. E. Green, W. Zerna, *Theoretical Elasticity*, 2nd ed., Oxford University Press, 1968.
- [4] E. G. Kirsch, Die Theorie der Elastizität und die Bedürfnisse der Festigkeitslehre, *Z. VDI* 42 (1898) 797–807.
- [5] R. C. J. Howland, On the stresses in the neighbourhood of a circular hole in a strip under tension, *Phil. Trans. R. Soc. A* 229 (1930) 49–86.
- [6] R. V. Southwell, On the analysis of experimental observations in problems of elastic stability, *Proc. R. Soc. A* 135 (1932) 601–616.
- [7] S. P. Timoshenko, J. N. Goodier, *Theory of Elasticity*, 3rd ed., McGraw-Hill, 1970.
- [8] L. R. Herrmann, Elasticity equations for incompressible and nearly incompressible materials by a variational theorem, *AIAA J.* 3 (1965) 1896–1900.
- [9] R. Hill, Elastic properties of reinforced solids: some theoretical principles, *J. Mech. Phys. Solids* 11 (1963) 357–372.

- [10] J.K. Knowles, Large amplitude oscillations of a tube of incompressible elastic material, *Quart. Appl. Math.* 18 (1960) 71–77.
- [11] R.D. Cook, Improved two-dimensional finite element, *J. Struct. Div. ASCE* 100 (1974) 1851–1863.
- [12] L. Anand, *Introduction to Coupled Theories in Solid Mechanics*, Oxford University Press, 2025; companion codes by E. Stewart and L. Anand, [solidmechanicscoupletheories.github.io](https://github.com/solidmechanicscoupletheories).
- [13] E.M. Arruda, M.C. Boyce, A three-dimensional constitutive model for the large stretch behavior of rubber elastic materials, *J. Mech. Phys. Solids* 41 (1993) 389–412.